

Integrated Case Management History and Analytics

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Abstract— Advanced case management gives enterprises the flexibility to organize their business operations in an efficient and transparent manner while providing knowledge workers the opportunity to proceed towards their goals and to react to exceptions as required in each individual situation. Powerful analytics based on pervasive observation of end-to-end case executions help to detect exceptional situations, determine possible derailment, and thus to prevent undesired outcomes. This paper demonstrates the application of proven technologies derived from real world examples to determine the health of a particular case.

I. INTRODUCTION

Business process management offers a wide array of tools and technologies to support efficient and effective business operations – from validation and simulation of process models at design-time, over application integration and workflow routing at runtime all the way to compliance and performance monitoring. While the benefits for well structured and highly repetitive processes are undeniable, traditional business process management lacks flexibility and agility in semi-structured, human-centric applications.

Case Management [1] represents a paradigm shift putting the goal and context of a business operation into the center of attention rather than the sequence of actions. Many vendors, including IBM [2], are putting out products that empower knowledge workers to choose the appropriate steps to execution, to interact and communicate naturally with their peers and leverage social networks, and to extend the scope and handling of a business case as dictated by a specific situation. Less predefined structure, however, opens the door to undetected exceptions, to determine possible derailment, and thus to undesired outcomes.

While enabling ad-hoc manipulations to predefined business operations, the challenge of advanced case management is to iteratively develop policies and best-practices guiding the behavior of all participants, to maintain oversight of the end-to-end case execution, and to analyze both execution traces and case data to improve the overall efficiency and/or to ensure regulatory compliance. In short, advanced case management attempts to transfer many of the tools and technologies from traditional BPM onto more flexible, faster deployable and more human friendly operation environments. As part of this effort, this paper presents ongoing research on *Case Health* – an assessment of the good standing of a particular case, based on a set of current measurements and forecasting analysis of expected case evolution. Seamlessly integrated into the case modelling and case worker environments, our continuous monitoring and predictive analysis of case health indicators provides early identification of ‘at risk’ cases, noti-

fies specified user(s) in real-time, and initiates or recommends remedial actions.

While the research is still ongoing and new and interesting questions arise faster than they could be answered, the core of the presented work has been integrated into a customer application with a major pharmaceutical supplier. Moreover, the ideas and solutions have been developed in close cooperation with the product development team of IBM Case Manager [3].

The remaining paper is structured as follows: In Section 2, we present an example and discuss manners of comprehensive case monitoring. Section 3 focuses on the visualization of execution traces on the appropriate level of abstraction for each different group of users. Section 4 addresses the tracking and prediction of Case Health and walks through the user experience. Section 5 discusses the architecture of the case management system, and Section 6 focuses on a case worker’s experience while using the system. In section 7 we conclude with avenues for future work.

II. CASE MONITORING

As less structure is imposed by a given case management system and the more flexibility and scope for action is given to a knowledge worker, it becomes exceedingly more important to maintain oversight on actual execution. Compiling much of our real world experiences, we use a fictitious company to illustrate this idea. Global Technology Delivery, Inc. (GTD) sells high-technology products to customers around the globe.

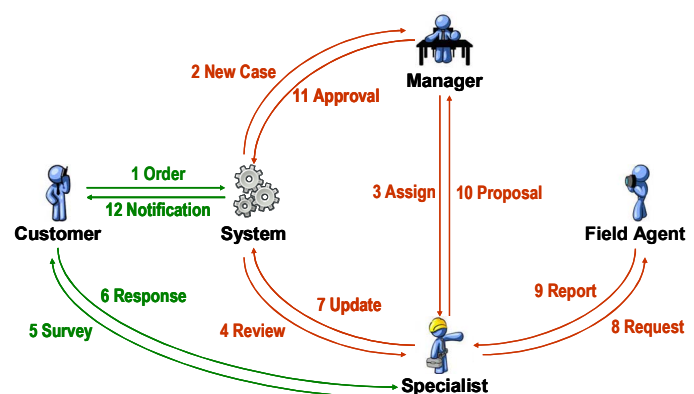


Figure 1 – Order Review Process at Global Technology Delivery, Inc.

The company has to ensure that all transaction comply with US export regulations. Each incoming order is automatically inspected with respect to a set of predefined red-flags. Once an order is marked suspicious, it is held pending a manual case evaluation. As shown in Figure 1, the specialist who the

case is assigned to has a variety of resources at his or her disposal to determine the legitimacy of the order.

The Order Review Process shown in Figure 1 is a typical case management solution combining mandatory, optional and ad-hoc steps as well as structured and unstructured data across multiple systems. With the order being flagged and the created case ID assigned, the specialist can access some data in the customer repository (content management), may contact the customer with a survey to collect more information (web application) or request the field agent to provide a report on the customer (via email). Additionally, he or she might upload documents into the case context. Parts of the specialist's duties are completely manual (read or write a report) while others are supported by snippets of automation (update customer profile or propose release of held order).

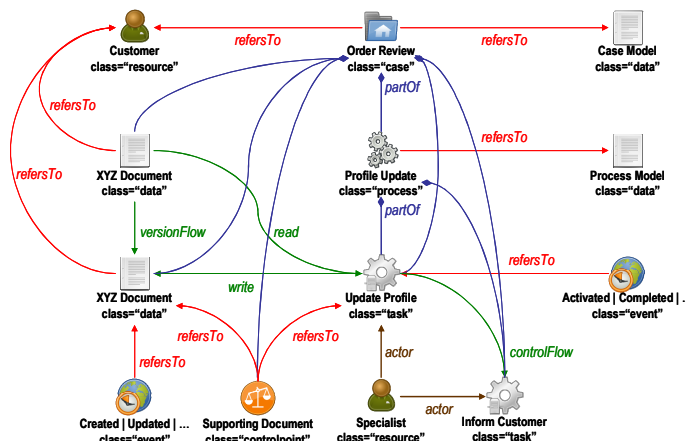


Figure 2 – A Partial Provenance Graph

Our goal is to gather a comprehensive end-to-end trace of each case execution without forcing the participants to manually declare their actions or to channel all their actions through an additional component. We use Business Provenance [4] to harvest events from the involved runtimes, connect those events with predefined or interactively mined correlation keys, and thus build a persistent and normalized representation of the execution sequence. We refer to this as the Provenance Graph [4]. Figure 2 shows a partial provenance graph for one case instance of our scenario.

A provenance graph consists of nodes representing relevant artifacts of different classes and edges representing semantic relations between those. Each node may carry a set of application dependent meta-data (e.g. case-id) and possibly a link to the real world artifact (e.g. url of the document version). As shown in Figure 2, each case instance is represented by a node with the link to the case model document. The case context consists of documents and performed tasks: Each document that is added or manipulated in the case has a corresponding node in the provenance graph that is linked to the case node. If a new document version is created, there will be a new provenance node as well with a *versionFlow* relation to the previous node. Each task that is executed in the case has corresponding node in the graph which is possibly linked to the

affected documents. If a task is part of a formal process, the process instance is also represented in the provenance graph.

The nodes for cases, processes, tasks and documents are created by correlation rules based on incoming events from the underlying runtime systems. As shown in Figure 3, gathered events concern documents, automated steps and state changes in external systems. In addition to instance specific nodes, the provenance graph represents application specific resources like the customer or the case specialist. Those nodes are typically linked to a multitude of instances and allow cross-case investigations.

With the upfront definitions of objects of interest in the case model, some nodes and edges can be created and linked automatically (i.e. through the case-id or customer-id). Other correlations are discovered in the recorded data based on unique values (key and foreign key relationships). In the end, each specific case solution might require additional manual definition of correlation rules – especially if compliance control points are to be tested automatically. In our experience, however, the cost of these efforts is a small fraction compared to the gain in productivity through easy access to case related artifacts, the potential liabilities of not being compliant with federal regulations (or not being able to prove compliance), or compared to a complete redesign and reimplementations of the order review process with a custom BPM application.

III. CASE HISTORY

While the provenance graph is a comprehensive representation of the end-to-end case execution traces, it is not the most effective means to communicate information to humans. With the complexity of today's business operations, in our opinion there is no feasible one-size-fits-all representation of the case history. Instead, the available information has to be filtered and projected for each distinct group of users in case management. Figure 3 shows five exemplary user groups.

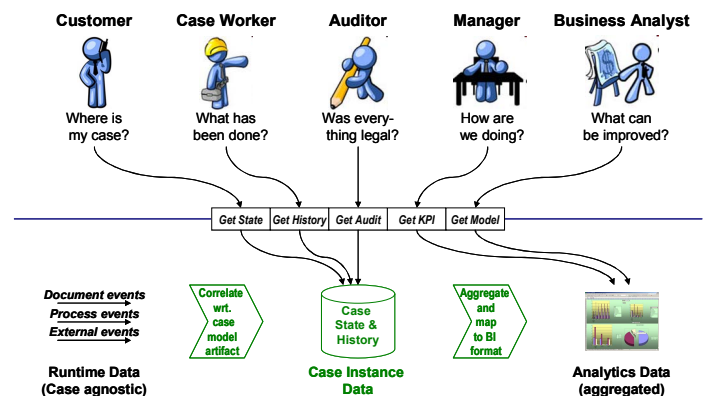


Figure 3 – Case History Gathering and Projection

Defining the right representation of the case execution history requires identifying the purpose served for each participant. With a case being handled by possibly many individuals, running for a long time or relying on external actions or events, the **case worker** needs to see clearly what has been accomplished in the case thus far. Our history projection provides a list of case relevant execution steps with the ability to

drill down into the details on who did what, when and where, and to retrieve the associated data. For the **case manager** such a representation holds an excessive number of irrelevant details and impedes a high level “big picture” view. Thus, our history projection provides a shorter list of major milestones that can be used to compute performance indicators and be aggregated for multiple case instances.

Even one step further back from the day-to-day operations, the **business analyst** receives access to generalized models which are mined from a multitude of case executions to showcase the actual behavior, bottlenecks and trends. With the case history being preserved in a normalized form, compliance audits are simple. The **auditor** has access to all instances to draw samples and can follow the execution in terms of control points. Moreover, many control points can be continuously monitored and thus be covered with 100% sample rate.

Finally, the **customer** requires a view into the status of their case. While it is obvious that neither detailed execution information nor compliance data is suitable for presentation in such a view, our history projection provides a list of externally visible milestones as part of a FedEx-like case tracking experience.

IV. CASE HEALTH

As mentioned before, the case history alone gives useful information and guidance to different stakeholders in the case management environment. Furthermore, it is the basis to determine how well each individual case instance is running so far and where it is likely to end up. We refer to this as Case Health. As shown in Figure 4, there are two different kinds of analytics for computing case health. Current case health is determined based on the case instance in question alone and yields results depending on events that have already occurred. Predictive case health on the other hand depends on a probabilistic model that has been derived from analytics conducted on aggregating many (previously executed) instances and is projected onto the case instance in question to predict events that might occur as the execution of the case proceeds [6].

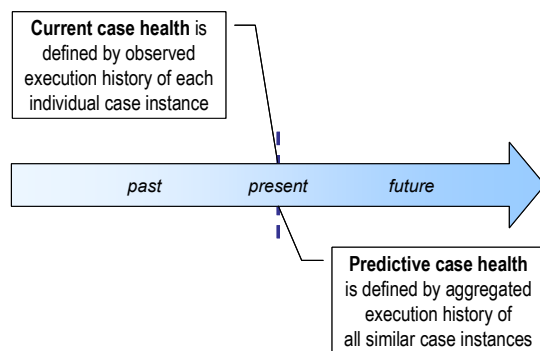


Figure 4 – Case Health Analytics

Indicators that constitute the overall case health depend on the individual case solution. In general, there are three classes of health indicators: performance indicators, potential for derailment indicators, and anomalous behavior indicators. Most indicators can be both detected in the execution so far as well

as predicted in the execution continuation. Each individual indicator is either discrete (normal, warning, alert) or has thresholds assigned to be mapped into discrete values. Figure 5 illustrates this concept.

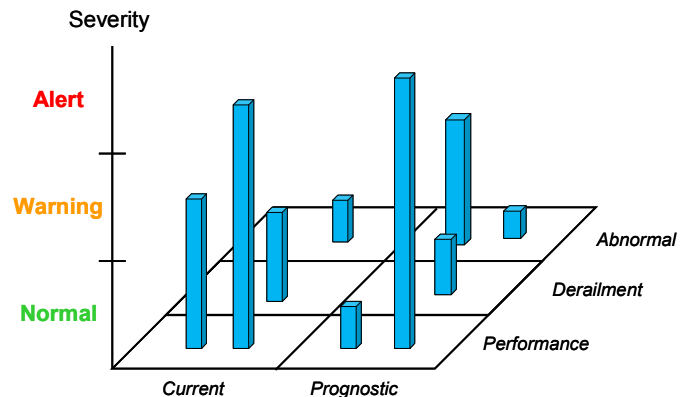


Figure 5 – Case Health Indicator Classes and Thresholds

A **performance indicator** refers to the timing of events, like a deadline, the duration, the idle-time or the count of repeating or reopening the particular task. When there is a deadline assigned to a task or a milestone, it seems simple to check if it has passed already. Since the deadline might depend on other events (e.g. respond with customer report to the request in 5 business days), current case health analytics use rules to determine the status of each deadline. Besides raising alerts in case of timer expiration, there are also warnings for deadlines being relatively near. With the knowledge derived from other instances, predictive analytics calculates the likelihood of missing a deadline. By aggregating case instance traces, a probabilistic graph can be built, where each node is a case outcome or state, and each edge represents the transition between two such states. If the graph is acyclic, simple application of the chain rule of conditional probabilities can allow prediction of the likelihood of any given outcome. Suppose our system has access to the data content values accessible to a case worker at a given decision point (i.e. a point where the execution branches) in such a graph. By applying a decision tree learning algorithm we can compute the correlation between the content of documents accessible to the case worker at that point and the execution of one of the descendent activities or next steps that the case worker can potentially take when handling the case. In other words, we can predict the outcome of an activity instance based on the contents of the documents accessible to a case worker when he or she executes that activity. For further details on how to configure and update a probabilistic graph for the purposes of predictive analytics with or without available data values, we refer the reader to our earlier work [6].

A **derailment indicator** refers to a situation that is legitimate with respect to the case solution but has been marked as undesirable, like the ending up in a bad situation (e.g. rejecting an order and thus losing business), skipping a critical task or missing a crucial document (which might have consequences in court), or any foreseen exception. While current

analytics can detect such situations to trigger remedial actions, predictive analytics has an even greater value by calculating the probabilities of derailment with respect to available choices, and thus giving guidance in making the right decisions.

In contrast, an **anomaly indicator** refers to a situation that lies outside of the legitimate behavior, such as a violation of a policy or the likelihood of a particular execution sequence being fraudulent. Current case health analytics based on rules cover the majority of such situations while predictive early warnings might be useful in select cases. In addition, an area of current research is to extract a notion of ‘normal’ behavior from the past case execution traces that can be fed into the analytics algorithms.

Depending on the severity level of each indicator, remedial action can be triggered from passive reporting on demand, over alerts sent via email or IM, to meetings being scheduled automatically, and finally as a last resort the halting of the current case. Our goal and topic of ongoing research is to make the specification of both health indicators and remedial actions as simple as possible and to provide intelligent defaults.

V. ARCHITECTURE

A Case Management system consists of three major components, as shown in Figure 6 below: the case builder to define the documents, tasks, roles and goals of the case execution, the case runtime to keep track of documents and integrate different enterprise applications, and the case client which provides the work environment for the case worker. Our work focuses on two additional components: the case history component providing access to end-to-end execution data for each case instance and the case analytics component that provides case health analytics capabilities,

The *Case History* component receives case-related events from a variety of sources, including the document management system, business process runtime and possibly other systems such as email repositories. Individual events (for example the case specialist uploaded a document into the document management system) need to be correlated with others to build a coherent trace that accurately represents the history of how a particular case was handled. This history is persisted in the form of a graph (the *provenance graph*) that stores the events, relationships between them, and includes inferences from low level events (such as inferring that a particular document update corresponds to completion of a milestone in the case). The provenance graph for any active case is continuously being transformed through the addition of new events and inferences. This process takes place within the Business Provenance system that underlies the case history component. The case history component provides a high level API to query the state of a case by inspecting the corresponding provenance graph.

The *Case Health* component consists, as described above, of two distinct pieces of functionality: support for current case health, and support for predictive case health.

Current case health is computed based on what has occurred in the case thus far; it is thus driven off case history

entirely. It is computed on demand and made available via the *Case Health Check Aggregator* when the user views the case health widget within the system user interface. As an example, consider the current health of a particular performance indicator, that a requested survey was received and uploaded to the document management system within 48 hours. When the user navigates to the case health widget in the case client, the case health component will execute the appropriate query on the case history component to determine if the survey was uploaded and, if so, whether it was done before the prescribed deadline. The widget immediately displays the current status of that health indicator to the user.

Predictive case health is more complex, and is supported by a number of specialized components. Firstly, completed cases retrieved from the case history are fed to a *Predictive Analytics Model Builder*. This is the training set used to drive the analytics, and results in the creation of a probabilistic process model and decision trees using methods outlined in the previous section and detailed in [6]. This information is combined with the real-time case status to produce a prediction at runtime. The case health component encapsulates this functionality and provides a simple API for the Case Management system to use it.

Consider the same performance indicator we discussed above. At any given state of the case, the predictive analytics can produce the likelihood that the deadline will be missed for this particular case. When the user navigates to the case health widget, this information is rapidly computed and retrieved for display via the mining and predictive analytics API.

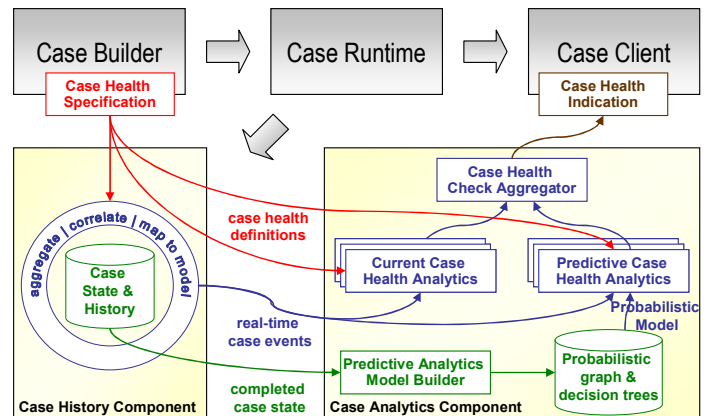


Figure 6 - Case History And Case Health Architecture

We have described so far how the system computes case history and case health and the various interactions that take place to present this information to the user. An additional aspect is the modeling time activities to configure case health. The case model is extended with health check constructs and the user specifies these when creating the model using the *Case Builder* widget. The case health definitions needed for a particular case are extracted from the model, mapped into appropriate queries and drive the measurement of both current and predictive case health.

VI. CASE STUDY

One major goal of case management is to empower the knowledge worker to decide which steps to take next. Take the scenario of a experienced nurse, the system should not stand in the way and complicate things but rather give flexibility and easy access, and take care of all the logging automatically. An inexperienced nurse, however, might be overwhelmed by the freedom and thus take the wrong decisions. Our case health approach can detect violations. Moreover, it draws on the experience of the cases handled before, and thus provides guidance. Finally, automatically recommended next steps from predictive analytics could be made subject to management approval to avoid potentially risky decisions. With respect to the scenario shown in Figure 1, Figure 7 displays a screenshot of the specialists ToDo-List.

Case Actions Case Documents Case Health		
ToDo List for current case		
Type	Description	Action
☐	Request Survey	Request additional information from Customer Request Survey
☐	Request Site Visit	Request additional information from Field Agent Request Site Visit
☒	Deny Order	Deny order if substantial doubts still remain after thorough investigation Deny Order
☒	Release Order	Release order if sufficient evidence for legitimate use was presented Release Order
Health Information		
☒	High probability of manager's rejection of Order Release Request.	
Health Information		
☒	Update Profile	Update customer profile Update Profile
Health Information		
☒	No recent customer site visit documentation to support customer profile update.	
☐	Add Document	Upload an additional document Add Document
☐	Add WorkItem	Add an additional workitem Add WorkItem

Figure 7 – Case Worker Experience

Figure 7 shows the ToDo-List of the specialist after she/he has opened an assigned case and updated the customer profile based on a News article found in the Financial Times. There are a total of seven possible next steps available. Three of those have a case health annotation next to it.

There is a case health alert next to “Update Profile” with the explanation: “No recent customer site visit documentation to support customer profile update.” This is an instance of a detected violation of a policy that has already occurred. The case worker has updated the customer profile based on the information found in the newspaper – momentarily not aware of a policy being in place to require a recent site visit report backing up the change. It was a conscious decision to not prevent the update itself but rather to raise an alert after the fact. Thus, it is possible to store some important information even without being able to get a report in time. As soon as such a report comes in, the specialist can make the alert disappear.

The yellow icons next to the two actions in Figure 7 are warnings that represent predictive analytics results. The specialist’s goal is to find enough evidence to decide either to release or to deny the order in question. The manager, however, has the final decision to approve the specialist’s request or to reject it. Obviously, rejecting the specialist’s request results in further investigation, and thus in an additional delay.

This is the reason why “rejecting the specialist’s request” is marked as an undesired outcome. With the currently available data, our analytics predict a high likelihood of the manager rejecting the specialist’s request. It seems unwise to proceed in this direction right now. Instead, the specialist could request a survey from the customer to broaden the knowledge available. With the additional data provided by the survey, the likelihood of the rejection drops under the threshold and the case health indicators turn green.

VII. CONCLUSION

Case management overcomes the rigidity of traditional BPM in situations of semi-structure, human centric applications. Effective case management requires the knowledge worker to be able to make informed decisions while being fully aware of the consequences. Our goal is to improve visibility into the state of a case by providing projections of case history tailored to role-specific needs (e.g. case worker, manager, customer, business analyst, and auditor) and to support measurement of health, performance and quality metrics using continuous monitoring.

To achieve this goal our case history component normalizes and correlates case-related events from different sources, provides predefined queries for role-based views and continuously audits, and monitors case health indicators. This approach is the backbone for our custom developed solution to review suspicious orders at a major pharmaceutical supplier.

Besides the already mentioned directions of current research, we pursue the topic of process mining for case management. While process mining [5] typically works on logs from a single (legacy) application, our goal is to utilize the case history and to use process-inspired models to summarize behavioural patterns found across historical case instances. By aggregating case traces, such a model reveals sequencing and decision logic patterns that might not be present in the original case model or that is only exposed by a few individual traces.

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